

Modelling and Parallelized Simulation of Natural Phenomena

N-Body Gravitational Simulation

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Contents

1	Introduction to the N-Body Problem	2
1.1	The N-Body problem	2
1.2	Historical Background	2
2	Special Cases with Analytical Solutions	3
2.1	The Two-Body Problem	3
2.2	Euler's Collinear Solutions	5
2.3	Lagrange's Triangular Solutions	6
2.4	Restricted Three-Body Problem	6
3	Numerical methods	6
3.1	Core Idea	6
3.2	Time Integration Methods	6
3.2.1	Taylor Expansion	7
3.2.2	Off-the-shelf Integrators	7
3.2.3	Symplectic Integrators	8
3.3	Force Calculation Methods	9
3.3.1	Direct Summation	9
3.3.2	Barnes-Hut Algorithm	9

1 Introduction to the N-Body Problem

1.1 The N-Body problem

The N-body problem is a fundamental problem in physics that involves predicting the future state(i.e. position and velocity) of each component in a system of N celestial bodies interacting with each other through gravitational forces. Now, this might seem like a straightforward problem, and much unrelated to computing, but that is very much not the case.

1.2 Historical Background

This first began in the 17th century, when Sir Issac Newton published his work "*Philosophiæ Naturalis Principia Mathematica*" in 1687 which laid the foundation for classical mechanics and provided the mathematical framework to describe the gravitational force between two bodies.

$$F = G \frac{m_1 m_2}{r^2} \quad (1)$$

Where:

- F is the gravitational force between the two bodies.
- G is the gravitational constant.
- m_1 and m_2 are the masses of the two bodies.
- r is the distance between the centers of the two bodies.

In theory, these equations could be used to turn the unpredictability of the universe to a deterministic system. Given a initial starting state of the objects, we can use the newton's gravitational formula, the laws of motion and the knowledge of calculus to predict the future state of the system at any time, in the future or in the past. However, in practical, we only have the exact solution(also known as analytic solution) for just the two-body problem. Adding even one more body to the system adds chaos to the motion of the bodies and makes it impossible to solve the equations analytically.

The N-body problem first came up when newton tried to apply the analytical two-body solution to the motion of planets in the solar system, after collecting the initial states. But over the course of a few years, he and others soon discovered that the motion of the planets was not exactly as predicted by the two-body solution. They realized that this was because gravitational interactive forces amongst all the planets were affecting all their orbits.

Since, they were able to solve the two-body problem easily, with the likes of kepler already having derived similar results(like the elliptical orbit) even before the formulation of gravity, the obvious next step was to solve the three boyd problem and the N-body problem. However, even after three centuries of research, there has been no practical analytical solution to the N-body problem other than a few specialized cases or conditions.

In the late 1800s, mathematicians Henri Poincaré and Ernst Burns asserted that no general solution exists for the N-body problem, and that the dynamics of the system is inherently chaotic, and depends sensitively on the initial conditions. This means that even

a small error in the initial state of the system can lead to vastly different outcomes over time, making long-term predictions extremely difficult. This was also one of the earliest discoveries of chaos theory, and a significant blow to the deterministic or the predictable view of the universe.

2 Special Cases with Analytical Solutions

While we do not have a practical analytical solution to the N-body problem, there has been several remarkable progress in the field with exact solutions for specialized cases, research about the analytical solutions and the development of numerical methods to approximate the solution. There are a few special cases of the N-body problem that have been solved analytically.

2.1 The Two-Body Problem

The two-body problem is the simplest non-trivial case of the N-body problem, where only two bodies are interacting with each other. This problem can be solved analytically using the law of universal gravitation, and langrangian mechanics.

Consider two point masses, m_1 and m_2 , located at positions $\mathbf{r}_1$ and $\mathbf{r}_2$, respectively. Let $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$ be the relative position vector from m_1 to m_2 . Then the force equation would be:

$$\mathbf{F} = -G \frac{m_1 m_2}{|\mathbf{r}|^3} \mathbf{r} \quad (2)$$

Now, take the Langrangian of the system, which is the difference between the kinetic energy and the potential energy:

$$L = T - U \quad (3)$$

Where:

- T is the kinetic energy of the system
- U is the potential energy of the system

Now let's try to calculate the kinetic energy of the system.

$$T = \frac{1}{2} m_1 |\dot{\mathbf{r}}_1|^2 + \frac{1}{2} m_2 |\dot{\mathbf{r}}_2|^2 \quad (4)$$

The problem is that we have two position vectors, giving us six degrees of freedom. So we need to reduce them. Let $\mathbf{R}$ be the center of mass of the system. Then:

$$\mathbf{R} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{M} \quad (5)$$

Where M is the total mass of the system and we know:

$$\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1 \quad (6)$$

Using (5) and (6), we can express $\mathbf{r}_1$ and $\mathbf{r}_2$ in terms of $\mathbf{R}$ and $\mathbf{r}$:

$$\mathbf{r}_1 = \mathbf{R} - \frac{m_2}{M} \mathbf{r} \quad (7)$$

$$\mathbf{r}_2 = \mathbf{R} + \frac{m_1}{M}\mathbf{r} \quad (8)$$

Substituting these into the expression for kinetic energy, we get:

$$T = \frac{1}{2}M|\dot{\mathbf{R}}|^2 + \frac{1}{2}\mu|\dot{\mathbf{r}}|^2 \quad (9)$$

Where μ is the reduced mass of the system, defined as:

$$\mu = \frac{m_1 m_2}{M} \quad (10)$$

Now take the Langrangian of the system using (3) substituting the expressions for kinetic energy:

$$L = \frac{1}{2}M|\dot{\mathbf{R}}|^2 + \frac{1}{2}\mu|\dot{\mathbf{r}}|^2 + U(r) \quad (11)$$

Now, we can apply the Euler-Lagrange equations to derive the equations of motion for the system. The Euler-Lagrange equations are given by:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0 \quad (12)$$

Where q_i are the generalized coordinates of the system. In our case, we can choose $\mathbf{R}$ as the generalized coordinates. Applying the Euler-Lagrange equations to $\mathbf{R}_x$ gives:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{R}_x} \right) - \frac{\partial L}{\partial R_x} = 0 \quad (13)$$

Now if we notice, L doesn't depend on $\mathbf{R}$, but only on $\dot{\mathbf{R}}$. So the second term of the above equation is zero. So we get:

$$\frac{d}{dt} (M\dot{R}_x) = 0 \quad (14)$$

This implies that $M\dot{R}_x$ is a constant, which means that the center of mass of the system moves with a constant velocity. So we can now choose the frame of reference of the center of mass, which means that $\dot{\mathbf{R}} = 0$, as we can later just add the constant velocity of the center of mass to the solution to get the solution in any frame of reference and similarly take $\mathbf{R}$ as zero too.

Now lets look at the angular momentum of the system. The angular momentum $\mathbf{L}$ is given by:

$$\mathbf{L} = \mathbf{r}_1 \times m_1 \dot{\mathbf{r}}_1 + \mathbf{r}_2 \times m_2 \dot{\mathbf{r}}_2 \quad (15)$$

Substituting the expressions for $\mathbf{r}_1$ and $\mathbf{r}_2$ in terms of $\mathbf{R}$ and $\mathbf{r}$, we get:

$$\mathbf{L} = \mu \mathbf{r} \times \dot{\mathbf{r}} \quad (16)$$

Now the angular momentum is conserved, which means that $\mathbf{L}$ is a constant. This implies that the motion of the two bodies is confined to a plane perpendicular to the angular momentum vector (One can show this easily by taking the time derivative of the angular momentum and showing that it is zero).

So let's take the plane of motion as the xy-plane, meaning that $\mathbf{r}$ can be written as $\mathbf{r} = (x, y, 0)$. So we are only left with two degrees of freedom. Now convert this to the

polar coordinates, where r is the distance between the two bodies and θ is the angle between the line connecting the two bodies and the x-axis. So we can write:

$$x = r \cos \theta \quad (17)$$

$$y = r \sin \theta \quad (18)$$

We can now express the kinetic energy in terms of r and θ :

$$T = \frac{1}{2}\mu(\dot{r}^2 + r^2\dot{\theta}^2) \quad (19)$$

Write the langrangian again, now in terms of r and θ :

$$L = \frac{1}{2}\mu(\dot{r}^2 + r^2\dot{\theta}^2) + U(r) \quad (20)$$

Now we can apply the Euler-Lagrange equations to derive the equations of motion for r and θ . Applying the Euler-Lagrange equations to θ gives:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0 \quad (21)$$

Now if we notice, L doesn't depend on θ , but only on $\dot{\theta}$. So the second term of the above equation is zero. So we get:

$$\frac{d}{dt} (\mu r^2 \dot{\theta}) = 0 \quad (22)$$

This implies that $\mu r^2 \dot{\theta}$ is a constant. We can denote this constant as l .

$$\mu r^2 \dot{\theta} = l \quad (23)$$

Now applying the Euler-Lagrange equations to r gives:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) - \frac{\partial L}{\partial r} = 0 \quad (24)$$

Substituting the expression for L , we get:

$$\frac{d}{dt} (\mu \dot{r}) - \left(\mu r \dot{\theta}^2 - \frac{dU}{dr} \right) = 0 \quad (25)$$

Substituting the expression for $\dot{\theta}$, we get:

$$\mu \ddot{r} - \frac{l^2}{\mu r^3} + \frac{dU}{dr} = 0 \quad (26)$$

Thus we got an equation with only one degree of freedom, which can be solved analytically to get the solution for r as a function of time.

2.2 Euler's Collinear Solutions

In 1767, Leonhard Euler discovered a special class of solutions to the three-body problem in which the three bodies remain on a single straight line at all times. The ratios of the distances between the bodies remain constant, preserving the shape of the configuration as it moves. These solutions are known as Euler's collinear solutions.

2.3 Lagrange's Triangular Solutions

Discovered by Joseph-Louis Lagrange in 1772, these solutions describe a configuration where three bodies form an equilateral triangle that rotates around the center of mass. The distances between the bodies remain constant, and the triangle maintains its shape as it moves through space. These solutions are known as Lagrange's triangular solutions.

2.4 Restricted Three-Body Problem

The restricted three-body problem is a special case of the three-body problem where one of the bodies has negligible mass compared to the other two. This allows for an analytical solution to be derived for the motion of the negligible mass body under the gravitational influence of the two larger bodies. This problem is often used to study the motion of satellites and spacecraft in the vicinity of two larger celestial bodies, such as the Earth and the Moon. While still being an approximation, it provides valuable insights into the dynamics of such systems and has practical applications in space exploration.

3 Numerical methods

Untill now, we only have the analytical solution to a few special cases of the N-body problem. For the general case, we have no practical solution to getting the accurate future state of these systems. Hence, these problems have to solved or more precisely simulated using numerical methods.

Numerical methods are techniques used to approximate the solutions of mathematical problems that cannot be solved, which can be either becuae we cannot solve the procedure analytically or because the analytical solution is intractable.

3.1 Core Idea

Solving the N-body problem numerically is basically a two-step process.

1. Calculate the forces acting on each body due to the gravitational attraction of all other bodies in the system.
2. Update the positions and velocities of each body based on the calculated forces using numerical integration.

Now depending upon the needs of the simulation, like the accuracy, computation resources available, the number of bodies in the system, and the time duration of the simulation, we have different methods to solve these steps.

Let's first look at the time integration methods, which are used to update the positions and velocities of the bodies based on the calculated forces.

3.2 Time Integration Methods

Firstly let's try to understand the problem and what we are trying to solve and find.

For a system of N bodies, let's say the position and velocity of the i^{th} body at time t are given by $\mathbf{r}_i^t$ and $\mathbf{v}_i^t$ respectively. The force acting on the i^{th} body due to the gravitational

attraction of all other bodies can be calculated using Newton's law of universal gravitation as:

$$\mathbf{F}_i^t = G \sum_{j \neq i} \frac{m_i m_j}{|\mathbf{r}_i^t - \mathbf{r}_j^t|^3} (\mathbf{r}_j^t - \mathbf{r}_i^t) \quad (27)$$

Where G is the gravitational constant, and m_i and m_j are the masses of the i^{th} and j^{th} bodies respectively. Rewriting it into differential form, we get:

$$\frac{d\mathbf{v}_i}{dt} = \frac{\mathbf{F}_i}{m_i} \quad (28)$$

$$\frac{d\mathbf{r}_i}{dt} = \mathbf{v}_i \quad (29)$$

Now we want to find the position and velocity of each body at a future time T . But since we don't know the exact solution to these equations, we try to move towards T in small time steps, say Δt , and update the positions and velocities of the bodies at each time step. Hence this process is called time integration.

3.2.1 Taylor Expansion

Now for calculating position (and similarly velocity) at a future time $t + \Delta t$, we can use the Taylor expansion of the position function $\mathbf{r}_i(t)$ around time t :

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \Delta t \frac{d\mathbf{r}_i}{dt} + \frac{(\Delta t)^2}{2!} \frac{d^2\mathbf{r}_i}{dt^2} + \frac{(\Delta t)^3}{3!} \frac{d^3\mathbf{r}_i}{dt^3} + \dots \quad (30)$$

Now, we can truncate this series after a certain number of terms to get an approximation of the position at time $t + \Delta t$. The more terms we include, the more accurate our approximation will be, but it will also require more computational resources. Truncating to the **n-th** term gives us an approximation of order n , known as the **n-th order method**.

Using this Taylor series, we can derive various time integration methods. These methods aren't just specifically used for the N-body problem, but are general methods used for solving ordinary differential equations and are very commonly used as off the shelf methods for solving such differential equations in various fields of science and engineering.

3.2.2 Off-the-shelf Integrators

There are various off-the-shelf integrators available for solving ordinary differential equations, which can be used for the N-body problem as well.

1. **Euler's Method:** This is the simplest numerical integration method, which uses the first-order Taylor expansion and approximates the curve as a straight line between two points.

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \Delta t \mathbf{v}_i(t) \quad (31)$$

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t) + \Delta t \mathbf{a}_i(t) \quad (32)$$

2. **Runge-Kutta Methods:** These are a family of higher-order methods that provide better accuracy by evaluating the function at multiple points within the time step.

- (a) **RK2**: Second order method, also known as the midpoint method.

$$k_1 = f(t, y) \quad (33)$$

$$k_2 = f\left(t + \frac{\Delta t}{2}, y + \frac{\Delta t}{2}k_1\right) \quad (34)$$

$$y(t + \Delta t) = y(t) + \Delta t k_2 \quad (35)$$

- (b) **RK4**: Fourth order method, which is the most commonly used Runge-Kutta method.

$$k_1 = f(t, y) \quad (36)$$

$$k_2 = f\left(t + \frac{\Delta t}{2}, y + \frac{\Delta t}{2}k_1\right) \quad (37)$$

$$k_3 = f\left(t + \frac{\Delta t}{2}, y + \frac{\Delta t}{2}k_2\right) \quad (38)$$

$$k_4 = f(t + \Delta t, y + \Delta t k_3) \quad (39)$$

$$y(t + \Delta t) = y(t) + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad (40)$$

While these methods already provide a good accuracy, they have a major disadvantage at the fact these equations do not conserve energy, as they are only trying to approximate the curve, without considering the underlying physics of the problem. This can lead to significant errors in long-term simulations, as the energy of the system can drift over time, leading to unphysical results.

3.2.3 Symplectic Integrators

Symplectic integrators are a class of numerical integration methods that are designed to preserve the phase-space structure of Hamiltonian systems, which includes the N-body problem. These integrators are particularly useful for long-term simulations of conservative systems, as they help to maintain the energy and other conserved quantities of the system over time. Symplectic integrators are based on the idea of splitting the Hamiltonian into separate parts, which can be integrated exactly or approximately, and then combining the results to obtain an overall update for the system.

Symplectic Integrators in itself is a specialized topic and is beyond the scope of this report and project and we'll be directly using some of the most commonly used symplectic integrators, the symplectic Euler, the Leapfrog and the Velocity Verlet integrators, which are both second-order methods.

1. **Symplectic Euler Integrator**: In normal Euler method, we first update the position using the old velocity, and then update the velocity using the old acceleration. This means that the position is updated without considering the inward acceleration which can lead to loss or increase in energy of the system. An easy way to fix this is to first update the velocity using the old acceleration, and then update the position using the new velocity. This is the symplectic Euler method.

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t) + \Delta t \mathbf{a}_i(t) \quad (41)$$

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \Delta t \mathbf{v}_i(t + \Delta t) \quad (42)$$

2. Leapfrog Integrator:

$$\mathbf{v}_i(t + \frac{\Delta t}{2}) = \mathbf{v}_i(t) + \frac{\Delta t}{2} \mathbf{a}_i(t) \quad (43)$$

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \Delta t \mathbf{v}_i(t + \frac{\Delta t}{2}) \quad (44)$$

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t + \frac{\Delta t}{2}) + \frac{\Delta t}{2} \mathbf{a}_i(t + \Delta t) \quad (45)$$

3. Velocity Verlet Integrator:

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \mathbf{v}_i(t) \Delta t + \frac{1}{2} \mathbf{a}_i(t) \Delta t^2 \quad (46)$$

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t) + \frac{1}{2} (\mathbf{a}_i(t) + \mathbf{a}_i(t + \Delta t)) \Delta t \quad (47)$$

3.3 Force Calculation Methods

The other step in the numerical simulation of the N-body problem is to calculate the forces acting on each body due to the gravitational attraction of all other bodies in the system. Now this can be done in many ways and is commonly the bottleneck in the simulation, as it requires calculating the forces between all pairs of bodies, which can be computationally expensive for large systems and brings in the diversity of methods to calculate the forces.

3.3.1 Direct Summation

The most straightforward method to calculate the forces is to directly sum the contributions from all other bodies for each body. This method has a computational complexity of $O(N^2)$, as we need to calculate the force between each pair of bodies. We can optimize this by using the fact that the force between two bodies is equal and opposite, so we only need to calculate the force for each pair once. However, this method is still computationally expensive for large systems, and is not practical for simulating systems with a large number of bodies.

3.3.2 Barnes-Hut Algorithm

The Barnes-Hut algorithm is a hierarchical method for approximating the forces in an N-body simulation. It works by dividing the simulation space into a hierarchical tree structure, where each node of the tree represents a region of space containing a group of bodies. The algorithm then approximates the forces from distant nodes by treating them as a single body located at the center of mass of the node. This allows for a significant reduction in the number of force calculations, as distant nodes can be approximated with fewer calculations. The Barnes-Hut algorithm has a computational complexity of $O(N \log N)$, making it much more efficient than direct summation for large systems.